

Organic Chemistry — Complete Notes

Grade 12

Physical Sciences

CAPS

What you'll learn

- Name organic compounds correctly using IUPAC rules.
- Recognise functional groups and place compounds in their homologous series.
- Write structural and condensed structural formulae.
- Identify chain, position, functional and geometric (cis/trans) isomers.
- Classify and balance addition, elimination, substitution, esterification and combustion reactions.
- Link physical properties such as boiling point and solubility to intermolecular forces.

What is organic chemistry?

Organic chemistry is the study of carbon compounds. Carbon is special because it forms four strong covalent bonds and can bond to other carbon atoms to make chains and rings of almost unlimited length. This allows millions of different compounds to exist, and they form the basis of fuels, plastics, medicines, foods and living tissue.

Most organic compounds contain carbon and hydrogen, and many also contain oxygen, nitrogen or a halogen (F, Cl, Br or I). In this topic you move from counting and naming molecules to predicting how they react.

Functional groups and homologous series

A functional group is an atom or group of atoms that gives a family of compounds its characteristic chemical reactions and its name. A homologous series is a family of compounds that share the same functional group and general formula, in which each member differs from the next by a $-\text{CH}_2-$ unit. Members of a homologous series show a gradual change in physical properties as the chain gets longer.

Homologous series	Functional group	General formula	Example
Alkanes	C–C single bonds only	$\text{C}_n\text{H}_{2n+2}$	methane CH_4
Alkenes	C=C double bond	C_nH_{2n}	ethene $\text{CH}_2=\text{CH}_2$
Alkynes	$\text{C}\equiv\text{C}$ triple bond	$\text{C}_n\text{H}_{2n-2}$	ethyne $\text{CH}\equiv\text{CH}$
Haloalkanes	C–X (X = F, Cl, Br, I)	$\text{C}_n\text{H}_{2n+1}\text{X}$	chloroethane $\text{CH}_3\text{CH}_2\text{Cl}$
Alcohols	–OH (hydroxyl)	$\text{C}_n\text{H}_{2n+1}\text{OH}$	ethanol $\text{CH}_3\text{CH}_2\text{OH}$
Aldehydes	–CHO (carbonyl at chain end)	R–CHO	ethanal CH_3CHO
Ketones	$>\text{C}=\text{O}$ (carbonyl mid-chain)	R–CO–R'	propanone CH_3COCH_3
Carboxylic acids	–COOH (carboxyl)	R–COOH	ethanoic acid CH_3COOH
Esters	–COO– (ester linkage)	R–COO–R'	ethyl ethanoate $\text{CH}_3\text{COOCH}_2\text{CH}_3$

The key difference between aldehydes and ketones is position: an aldehyde's carbonyl group sits at the end of the chain and always has an H attached to it, while a ketone's carbonyl group sits inside the chain between two carbon atoms.

IUPAC naming rules

Follow these steps in order:

1. Find the longest continuous carbon chain; this is the parent chain and gives the stem name (meth-, eth-, prop-, but-, pent- and so on).
2. Identify the functional group and attach the correct suffix.
3. Number the chain so that the functional group and then the substituents get the lowest possible position numbers.
4. Name and number any branches (substituents), listing them alphabetically.
5. Use hyphens between numbers and words (for example, propan-1-ol) and commas between separate numbers (for example, 2,3-dimethylbutane).

Common suffixes: -ane (alkane), -ene (alkene), -yne (alkyne), -ol (alcohol), -al (aldehyde), -one (ketone), -oic acid (carboxylic acid). Esters are named as two words: the alcohol part first (as an -yl) and the acid part second (as an -oate).

Halogen substituents are named as prefixes: fluoro-, chloro-, bromo- and iodo-.

Condensed formula	IUPAC name
$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$	propan-1-ol
$\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$	propan-2-ol
$\text{CH}_3\text{CH}=\text{CHCH}_3$	but-2-ene
$\text{CH}_3\text{CH}_2\text{COOH}$	propanoic acid
$\text{CH}_3\text{COOCH}_2\text{CH}_3$	ethyl ethanoate
$\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$	butanal
$\text{CH}_3\text{COCH}_2\text{CH}_3$	butan-2-one
$\text{CH}_2\text{ClCH}_2\text{CH}_3$	1-chloropropane

Isomerism

Isomers are compounds that have the same molecular formula but different structural formulae. Because they have different arrangements of atoms, they usually have different physical properties and sometimes different chemical properties.

Structural isomers

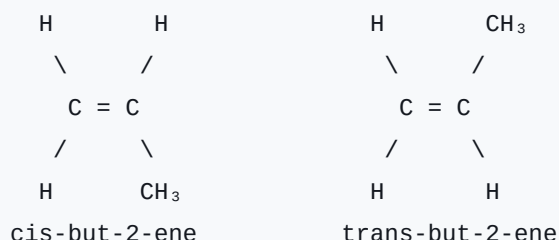
Structural isomers have the same molecular formula but the atoms are joined in a different order. There are three types you must know:

- Chain isomers: the carbon chains are branched differently. Example: butane, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$, and methylpropane, $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3$, both have the formula C_4H_{10} .
- Position isomers: the same functional group sits at a different position on the chain. Example: propan-1-ol and propan-2-ol, both $\text{C}_3\text{H}_8\text{O}$.
- Functional group isomers: the same molecular formula but a different functional group. Example: propanal, $\text{CH}_3\text{CH}_2\text{CHO}$, and propanone, CH_3COCH_3 , both $\text{C}_3\text{H}_6\text{O}$.

Geometric isomers (cis/trans)

Geometric isomers occur only when rotation is restricted and each carbon of the restricted bond carries two different groups. In this course, that means a carbon–carbon double bond (C=C) where each doubly bonded carbon is attached to two different atoms or groups.

- cis isomer: the two matching groups are on the same side of the double bond.
- trans isomer: the two matching groups are on opposite sides of the double bond.



If either doubly bonded carbon carries two identical groups, cis/trans isomers are not possible.

Reaction types

Organic reactions in Grade 12 fall into four main families, plus combustion.

Addition reactions (alkenes and alkynes)

Atoms add across the double or triple bond, turning an unsaturated compound into a saturated one. The number of carbon atoms never changes.

- Hydrogenation: $\text{alkene} + \text{H}_2 \rightarrow \text{alkane}$ (catalyst: Pt or Ni).
- Hydration: $\text{alkene} + \text{H}_2\text{O} \rightarrow \text{alcohol}$.
- Halogenation: $\text{alkene} + \text{Br}_2 \rightarrow \text{dibromoalkane}$. This is how bromine water is used to test for a double bond — the orange-brown colour disappears as the bromine adds across the C=C.
- Hydrohalogenation: $\text{alkene} + \text{HX} \rightarrow \text{haloalkane}$ (X = Cl, Br, I).

Elimination reactions

Elimination is the reverse of addition: atoms are removed from two adjacent carbon atoms, forming a double bond and a small molecule such as water or hydrogen halide.

- Dehydration of an alcohol: $\text{alcohol} \rightarrow \text{alkene} + \text{water}$ (conc. H_2SO_4 or hot Al_2O_3 as catalyst).
- Dehydrohalogenation of a haloalkane: $\text{haloalkane} + \text{strong base} \rightarrow \text{alkene} + \text{HX}$.

Substitution reactions

One atom or group is replaced by another; the carbon skeleton stays the same.

- Haloalkane + NaOH (aqueous) $\rightarrow$ alcohol + NaX.
- Alkane + halogen (in ultraviolet light) $\rightarrow$ haloalkane + HX.

Esterification

A carboxylic acid reacts with an alcohol to form an ester and water. Concentrated sulfuric acid is used as a catalyst, and the reaction is reversible.



Combustion

All organic compounds burn in oxygen. Complete combustion produces carbon dioxide and water. In a limited supply of oxygen, combustion is incomplete and produces carbon monoxide (a poisonous gas) and/or carbon (soot).

Physical properties and intermolecular forces

- Boiling point increases down a homologous series because longer chains have a larger surface area, so the London (van der Waals) forces between molecules are stronger.
- Branching lowers the boiling point: a branched molecule is more compact, has less surface contact, and so has weaker intermolecular forces than its straight-chain isomer.
- Alcohols and carboxylic acids have much higher boiling points than alkanes of similar mass because they form hydrogen bonds between molecules.
- Solubility follows "like dissolves like". Small alcohols dissolve in water because they hydrogen-bond with water; as the chain grows, the non-polar part dominates and solubility falls.

Worked example 1: Naming a branched compound

Name the compound $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_3$.

Step 1 — Find the longest chain. The longest continuous carbon chain has four carbon atoms, so the parent is butane.

Step 2 — Number the chain from the end that gives the branch the lowest number. Numbering from the left puts the methyl group on carbon 2; numbering from the right puts it on carbon 3. Choose carbon 2.

Step 3 — Name the branch and combine. One methyl group on carbon 2 gives: 2-methylbutane.

Worked example 2: Esterification

Ethanol reacts with ethanoic acid. Write a balanced equation and name the ester formed.

Step 1 — Write the acid and alcohol. CH_3COOH (ethanoic acid) + $\text{CH}_3\text{CH}_2\text{OH}$ (ethanol)

Step 2 — Combine them, losing one water molecule (the $-\text{OH}$ from the acid and an $-\text{H}$ from the alcohol). Add the catalyst over the arrow. $\text{CH}_3\text{COOH} + \text{CH}_3\text{CH}_2\text{OH} \rightarrow \text{CH}_3\text{COOCH}_2\text{CH}_3 + \text{H}_2\text{O}$ (conc. H_2SO_4)

Step 3 — Name the ester. The alcohol part is the ethyl group and the acid part gives ethanoate, so the product is ethyl ethanoate.

Worked example 3: Combustion and isomers

(a) Write the balanced equation for the complete combustion of ethanol. (b) How many structural isomers have the molecular formula C_4H_{10} ?

Step 1 — Write the unbalanced equation. $\text{C}_2\text{H}_5\text{OH} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$

Step 2 — Balance carbon, then hydrogen, then oxygen. There are 2 C and 6 H on the left, so place 2CO_2 and $3\text{H}_2\text{O}$ on the right. $\text{C}_2\text{H}_5\text{OH} + \text{O}_2 \rightarrow 2\text{CO}_2 + 3\text{H}_2\text{O}$

The right-hand side now has $4 + 3 = 7$ oxygen atoms. Ethanol supplies 1, so 6 come from O_2 , meaning 3 O_2 molecules. $\text{C}_2\text{H}_5\text{OH} + 3\text{O}_2 \rightarrow 2\text{CO}_2 + 3\text{H}_2\text{O}$

Check: C = 2 on both sides; H = 6 on both sides; O = 1 + 6 = 7 on the left and 4 + 3 = 7 on the right.
Balanced.

Step 3 — For part (b), write the straight chain and then all branched arrangements with four carbons.

- butane: $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
- 2-methylpropane: $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3$

There are 2 structural isomers of C_4H_{10} .

Common mistakes

- Failing to find the longest chain, which gives the wrong stem name.
- Numbering the chain from the wrong end so the functional group or branch gets the wrong position.
- Writing "1-propanol" by mistake when the functional group can only be at one position (correct: propan-1-ol is fine because there is a choice; but for ethanol it is simply ethanol, not 1-ethanol).
- Confusing aldehydes ($-\text{CHO}$, at the chain end) with ketones ($>\text{C}=\text{O}$, mid-chain).
- Thinking every alkene can show cis/trans isomerism — it must have two different groups on each doubly bonded carbon.
- Mixing up addition (an alkene gaining atoms across a double bond) with substitution (an atom being replaced).
- Forgetting that water is produced in esterification and in the dehydration of an alcohol.
- Drawing molecular formulae when a structural formula was requested.
- Forgetting the concentrated H_2SO_4 catalyst and that esterification is reversible.

Quick summary / cheat sheet

Homologous series -> suffix

alkane	-ane	$\text{C}_n\text{H}_{2n+2}$	
alkene	-ene	C_nH_{2n}	(has $\text{C}=\text{C}$)
alkyne	-yne	$\text{C}_n\text{H}_{2n-2}$	(has $\text{C}\equiv\text{C}$)
alcohol	-ol	$\text{C}_n\text{H}_{2n+1}\text{OH}$	(-OH)
aldehyde	-al	$\text{R}-\text{CHO}$	($\text{C}=\text{O}$ at end)
ketone	-one	$\text{R}-\text{CO}-\text{R}'$	($\text{C}=\text{O}$ mid-chain)
acid	-oic acid	$\text{R}-\text{COOH}$	
ester	...yl ...oate	$\text{R}-\text{COO}-\text{R}'$	
haloalkane	fluoro-/chloro-/bromo-/iodo-		

Reactions:

Addition	(alkene)	$\text{C}=\text{C}$ opens, atoms add across
Elimination	(alcohol/haloalkane)	double bond forms, small molecule lost
Substitution	(alkane/haloalkane)	one group replaced
Esterification	acid + alcohol	-> ester + water (conc. H_2SO_4)
Combustion	+ O_2	-> CO_2 + H_2O (complete)

Practice questions

1. Give the IUPAC name of $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$.
2. Draw and name all the structural isomers with the molecular formula C_4H_{10} .
3. Write a balanced equation for the complete combustion of methane (CH_4).
4. Ethanoic acid reacts with methanol. Name the ester formed and write a balanced equation.
5. Classify each reaction as addition, elimination or substitution:
(a) ethene + $\text{Br}_2 \rightarrow$ 1,2-dibromoethane (b) ethanol $\rightarrow$ ethene + water (c) propane + Cl_2 in ultraviolet light $\rightarrow$ chloropropane + HCl
1. Explain why the boiling point of pentane is higher than that of 2,2-dimethylpropane, even though both have the molecular formula C_5H_{12} .
2. Bromine water is shaken with an alkene and with an alkane. State and explain what is observed in each case.

Answers

1. butan-1-ol (the $-\text{OH}$ is on the first carbon of a four-carbon chain).
2. butane, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$, and 2-methylpropane, $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3$. That is 2 isomers.
3. $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$. Carbon balances (1 each), hydrogen balances (4 each) and oxygen balances (4 each).
4. The ester is methyl ethanoate: $\text{CH}_3\text{COOH} + \text{CH}_3\text{OH} \rightarrow \text{CH}_3\text{COOCH}_3 + \text{H}_2\text{O}$ (with concentrated H_2SO_4 as catalyst).
5. (a) addition, because atoms add across the $\text{C}=\text{C}$ of the alkene; (b) elimination, because a double bond forms and water is lost; (c) substitution, because a hydrogen atom on the alkane is replaced by a chlorine atom.
6. Pentane is a straight-chain molecule, so its molecules can pack closely and have a large contact surface area, giving stronger London forces. 2,2-dimethylpropane is more branched and compact, so its molecules have less contact area and weaker London forces. Weaker forces mean a lower boiling point for the branched isomer.
7. With the alkene the bromine water rapidly loses its orange-brown colour, because bromine adds across the $\text{C}=\text{C}$ in an addition reaction. With the alkane there is no reaction in the dark and no colour change, because an alkane is saturated and has no double bond to add across.

Official EduRank Team notes. Always cross-check with your textbook and teacher.